Quiz 9

Friday, November 6 RREF and one-step dynamics (Lecture 22)

Last updated: November 4, 2026.

Textbook: Boyd & Vandenberghe, *VMLS* (PDF)

Covers Lecture 22 only: reading RREF (pivot / free variables; unique / none / infinitely many) and one-step linear dynamics $x_{t+1} = Ax_t$ (build A, compute x_1). Same-day determinants and Lab 4 are **not** on this quiz. Matrix products / powers (Lecture 23) and Lab 3 are **not** on this quiz.

Schedule

Core

Fluency

C9.1. The RREF of an augmented matrix is

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & 3 \\ 0 & 1 & -1 & 4 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

(a) Which variables are pivots? Which are free?
(b) Write the general solution $x = (x_1, x_2, x_3)$ in terms of a parameter t.

C9.2. Classify each RREF as **unique**, **none**, or **infinitely many** solutions. One short reason each.

(a)
$$\left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & -1 \end{array}\right]$$

(b)
$$\left[\begin{array}{cc|c} 1 & 2 & 3 \\ 0 & 0 & 1 \end{array}\right]$$

(c)
$$\left[\begin{array}{ccc|c} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 4 \end{array}\right]$$

C9.3. From the RREF

$$\left[\begin{array}{ccc|c} 1 & 3 & 0 & 2 \\ 0 & 0 & 1 & -1 \end{array}\right],$$

name the free variable and write $x = (x_1, x_2, x_3)$ with a parameter.

C9.4. True or false, with a one-line reason: if RREF has a row of all zeros **including** the right-hand side, the system is inconsistent.

C9.5. A consistent RREF of $[A \mid b]$ for a 3×5 matrix A has 3 pivots. How many free variables? Unique solution or infinitely many?

Meaning / dynamics

C9.6. Two towns, Arcata and Eureka, exchange residents each year. Order $x_t = (\text{Arcata}, \text{Eureka})$.

- Of those in Arcata, 80% stay and 20% move to Eureka.
- Of those in Eureka, 90% stay and 10% move to Arcata.

(a) Write the transition matrix A so that $x_{t+1} = Ax_t$.

(b) With $x_0 = (5000, 7000)$, compute x_1. Does Arcata s population rise or fall that year?

C9.7. True or false, with a one-line reason: if $x_{t+1} = Ax_t$, then knowing x_t and A determines x_{t+1} uniquely.

C9.8. Two warehouses, $x_t = (\text{A}, \text{B})$. Each day, A keeps 70% and ships 30% to B; B keeps 60% and ships 40% to A.

(a) Write A.

(b) With $x_0 = (100, 50)$, find x_1.

C9.9. With $A = \begin{bmatrix} 1.1 & 0 \\ 0 & 0.5 \end{bmatrix}$ and $x_0 = (2, 8)$, compute x_1 and x_2. Which component is growing? Which is shrinking?

C9.10. You solve $Ax = b$ in software and get $x = (1, -1)$. You compute $Ax - b = (0.0,\ 10^{-15})$. Was the solve successful? Why check the residual? (One or two sentences.)

Textbook Core

- Boyd 9.1 9.3 (linear dynamical systems); review RREF ideas from class notes / Lecture 22.

Stretch

S9.1. Three age groups (youth, adult, senior):

$$A = \begin{bmatrix} 0.6 & 0 & 0 \\ 0.4 & 0.7 & 0 \\ 0 & 0.3 & 0.8 \end{bmatrix}, \qquad x_0 = (100, 200, 50).$$

Interpret A_{21} and A_{33} in one phrase each, and compute x_1.

S9.2. From

$$\left[\begin{array}{ccc|c} 1 & 0 & -2 & 5 \\ 0 & 1 & 3 & -1 \\ 0 & 0 & 0 & 0 \end{array}\right],$$

list pivot variables, free variables, and the general solution (same skill as C9.1 with different numbers).

What a quiz item might change

The RREF entries, which variable is free, the stay/move percentages in a two-compartment story, or the starting state x_0.